

The Anti-Reflective Properties of Subwavelength Structures on Lenses

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Abstract

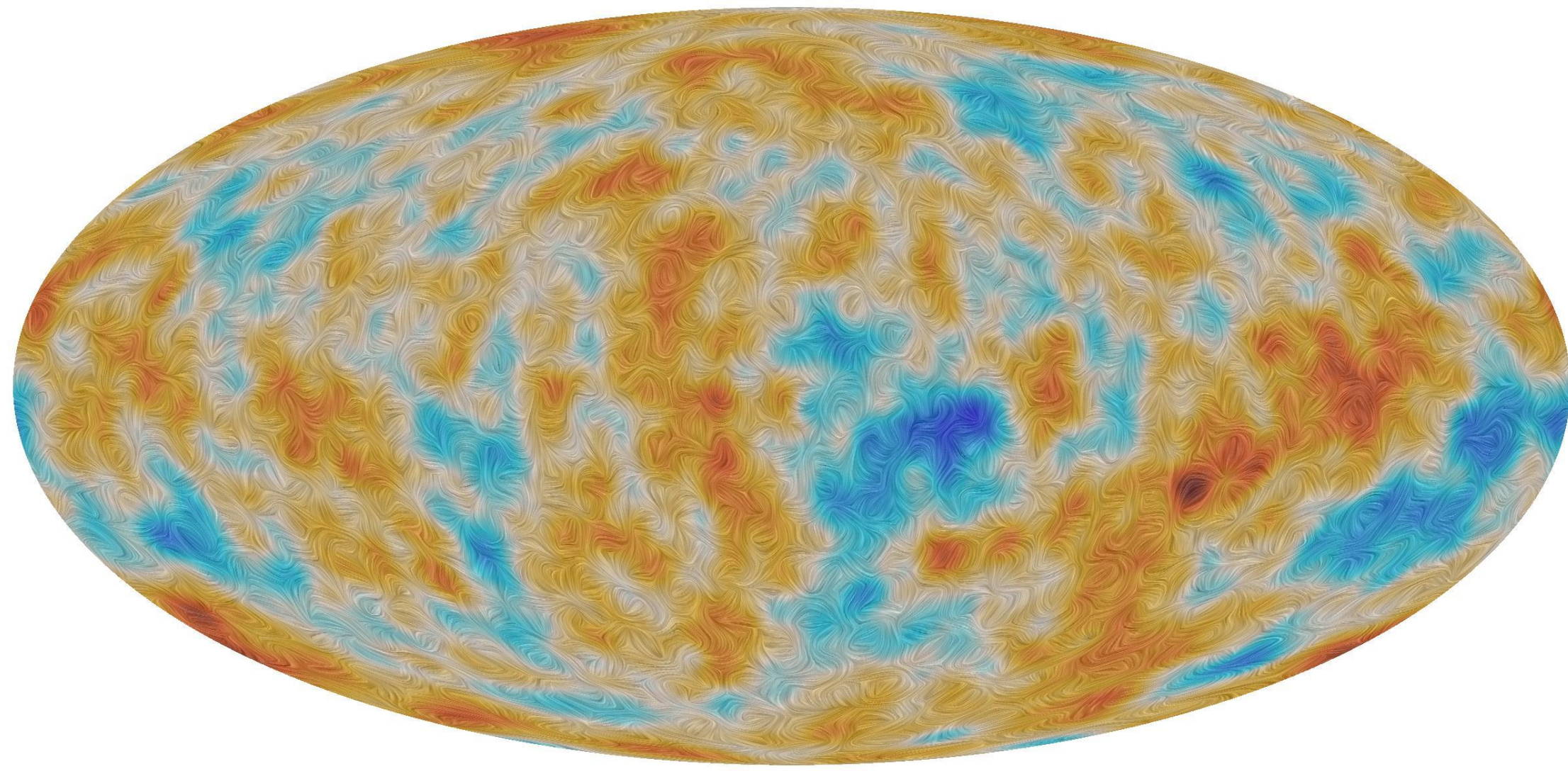


Figure 1: Polarization of the cosmic microwave background radiation

The Cosmic Microwave Background (CMB) radiation is the oldest detectable light in our universe, imprinted on the sky when the Universe was just 380,000 years old.^[1] Produced during the creation of the universe, a fraction of this radiation is slightly polarized – vibrating in preferred directions. The study of this polarization gives physicists an insight into the distribution of energy and matter in the early universe. To study this polarization with precision, astronomers need sensitive equipment which allow the maximum amount of light to be collected during observation. As a result, numerous anti-reflective coating techniques have been developed, including laser ablated subwavelength structures (SWS). This poster will include the process for simulating and modeling the transmission of light through subwavelength structures on alumina lenses.

Theory

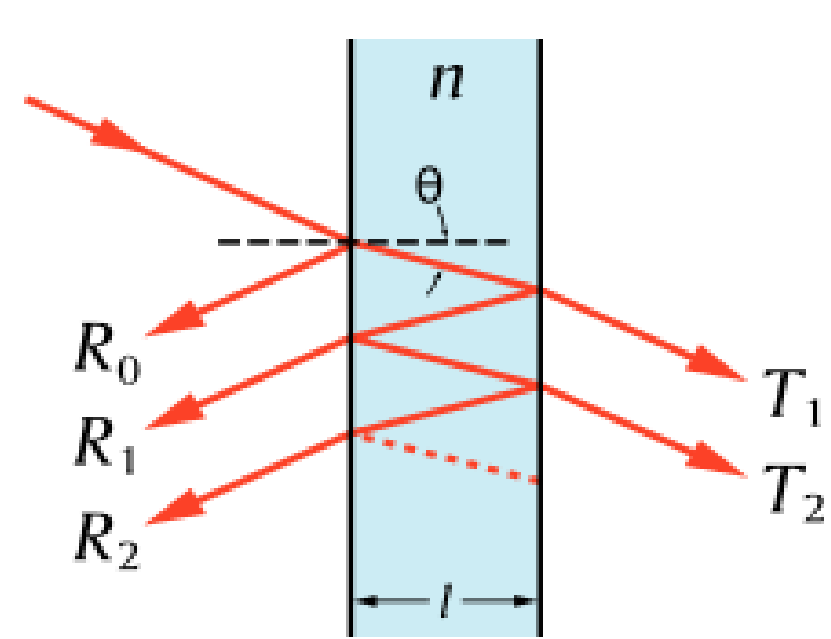


Figure 2: Simplified visual representation of light as it propagates through a material (left)

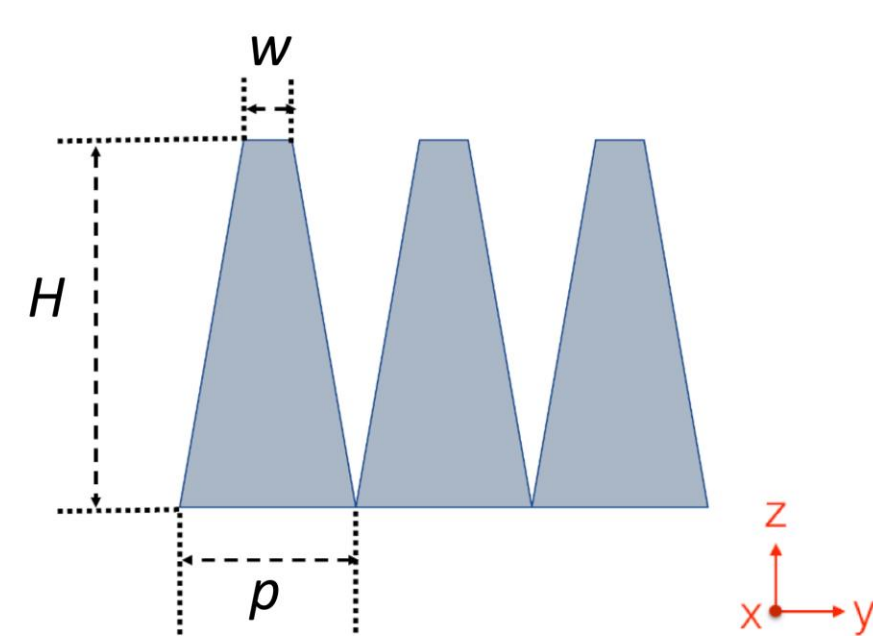


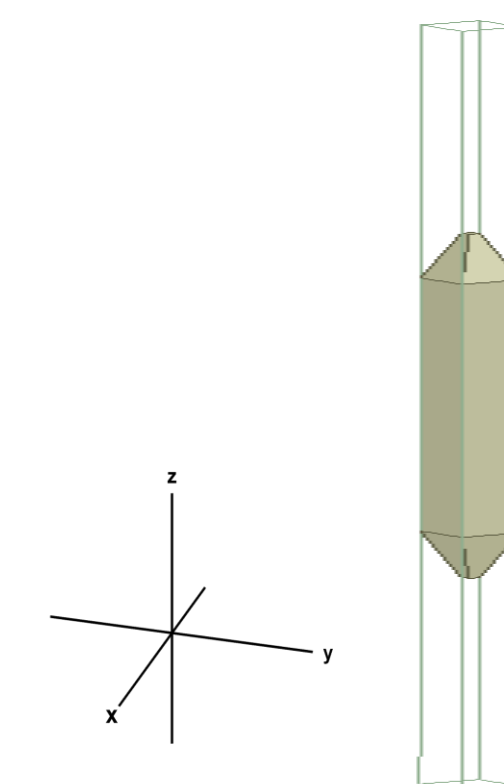
Figure 3: Side profile of SWS. Scale is on the order of microns (right)

- As electromagnetic energy propagates through a given material, energy can be reflected, absorbed, or transmitted through the material
- Alumina's high thermal conductivity and transmission spectrum make it an appealing low pass filter material for instruments operating at millimeter wavelengths^[2]
- One of the challenges in using alumina as a filter material is its tendency to reflect light due to the materials high index of refraction $n = 3.12$ ^[3]
- One technique of mitigating reflection is achieved by fabricating anti-reflective subwavelength structures on the surface of the alumina lens through laser ablation

Methodology

- Anslys High Frequency Structure Simulator (HFSS) is a simulation tool which allows for the study of electromagnetic energy using finite element analysis (FEA)
- The transfer matrix method (TMM) is a method is used in optics to analyze the propagation of electromagnetic waves through a medium.^[4] Using TMM, simulations were coded which represented that of an alumina lens with infinite radius coupled with SWS anti-reflective coating (ARC).
- Using Anslys HFSS, a partial element (including SWS ARC) of the entire lens can be modeled (figure 4). Infinite periodic boundary conditions can then be applied to the partial element to represent an infinite lens with SWS. Results comparing the two methods can therefore be analyzed

Figure 4: HFSS model of partial element of alumina lens with pyramid shaped SWS placed above and below the substrate



Results

- Transmission was predicted using Anslys HFSS of a subwavelength structure in the 0 THz to 1 THz frequency range
- HFSS modeling verifies anti-reflective properties of SWS, enabling the prediction of complex surfaces with SWS

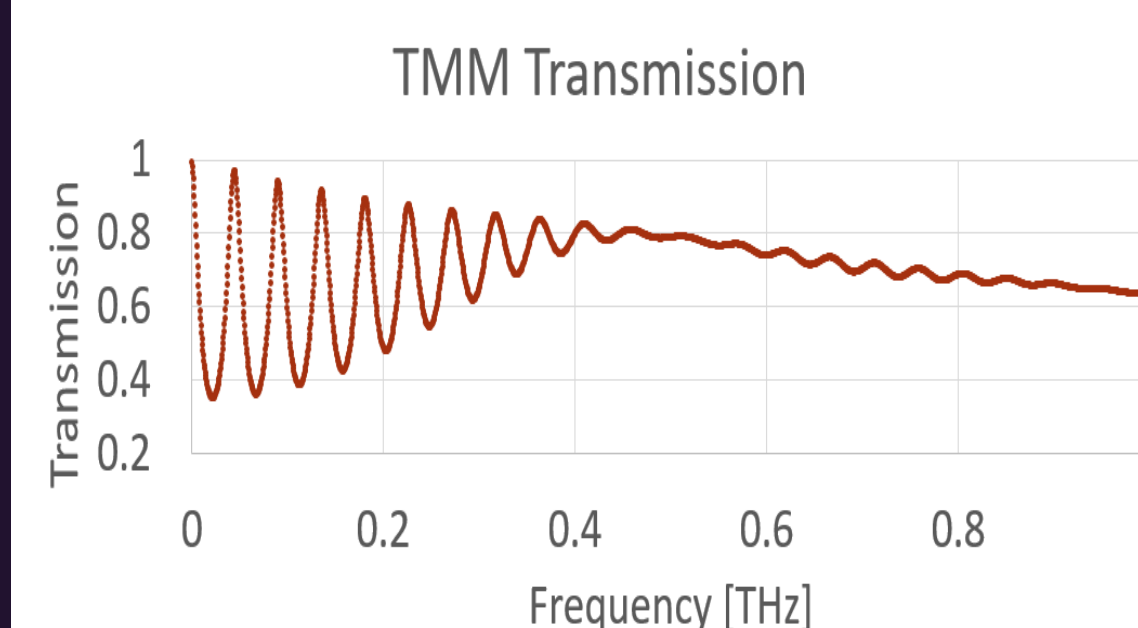


Figure 5: Transmission predicted using HFSS FEA method (above)

Figure 7: Difference in predicted transmission between HFSS and TMM. Results agree to within 3%

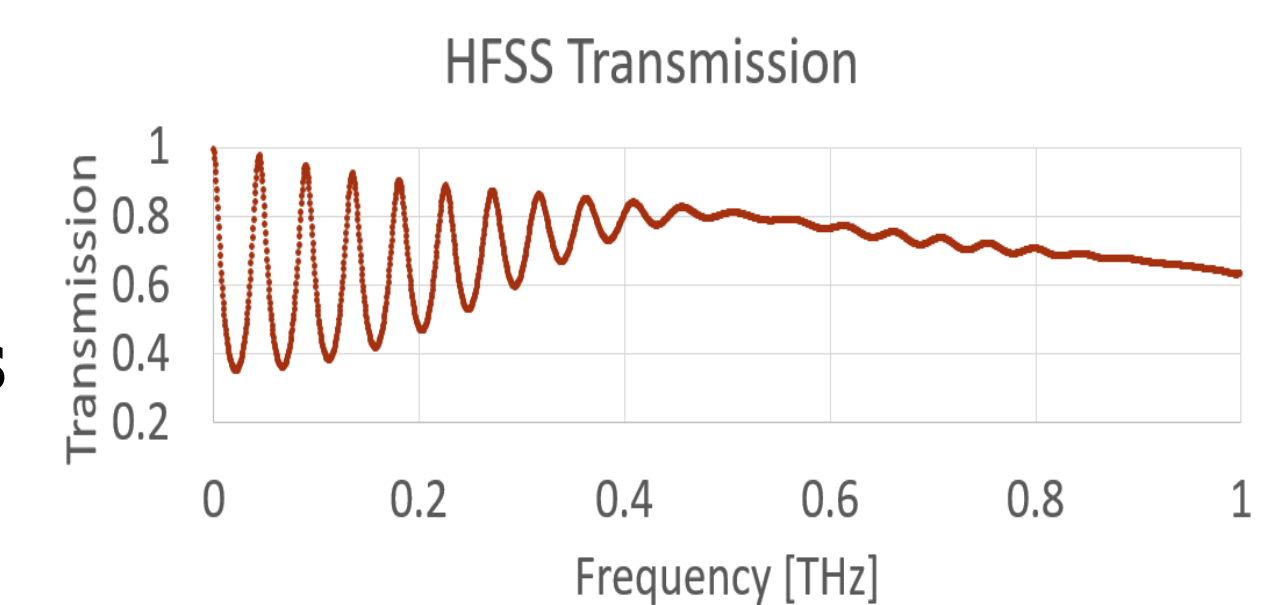


Figure 6: Transmission predicted by analytical TMM model (above)

Acknowledgements

[1] NASA
[2] Y. Inoue, T. Matsumura, M. Hazumi, A. T. Lee, T. Okamura, A. Suzuki, T. Tomaru, and H. Yamaguchi, "Cryogenic infrared filter made of alumina for use at millimeter wavelength," Appl. Opt. 53, 1727-1733 (2014)
[3] J. W. Lamb, "Miscellaneous data on materials for millimeter and submillimeter optics," Int. J. Infrared Millim. Waves 17, 1997-2034 (1996)
[4] Born, M.; Wolf, E. *Principles of Optics: Electromagnetic Theory of Propagation, Interference, and Diffraction of Light*. Oxford, Pergamon Press, 1964

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